



Descriptive Statistics

Key terms

KSBs: K13, S11, S52 B1

Descriptive statistics, measures of location, central tendency, mean, median, mode, measures of spread, variance, standard deviation, discrete data, continuous data, categorical data

2.1 Types of Data

Data comes in many different types, and we can categorise it by the number of variables involved.

Definition 2.1 - Data types by number of variables

Univariate data is data with only *one variable*. If we have *two variables* and want to study the relationship between them, we say the data is **bivariate**. We use the term **multivariate** for data with *more than one variable*.

Some data is *qualitative*, like categories we can classify things into. For example, *hair colour* is a qualitative data type, since the outcomes could be *blonde, ginger, black*. etc. Other data is *quantitative* which means it takes *numerical values*.

Definition 2.2 - Discrete and continuous data

Discrete data is numerical data which can take *one of a fixed set of values* like the number of *objects in a set*, or *"age in full years"*.

Continuous data can take values *anywhere on a scale* like a measurement of *length or time*.

We will mostly be considering numerical data with a single variable, although the tools we are discussing today can also be applied to multivariate data.

Descriptive statistics allow us to describe sets of numerical data and their properties, without including the full data set. They can be used to summarise what data is telling us, and compare different sets of data.

2.2 Measures of Location

These are ways to describe **where** numerical data is, within the range of possible values it can take. One type that is used commonly is a **measure of central tendency**, which gives an indication of where the *centre of the data is, for some definition of "centre"*.

2.2.1 Mean

The mean is the most commonly used type of **average**, and is *the value you would expect given the data.*

Definition 2.3 - Mean

The **mean** of a set of values is an average found by taking the *sum of all the values in the set* and dividing by *the number of values.*

For a set of n values, we write $\sum x$ for the sum of the values, and the mean is $\frac{\sum x}{n}$. The mean for a sample is denoted $\bar{x}$ ("x bar")

Example 2.1.

Consider the set of values:

4 9 7 5 1 8 3 4 9 5 4

The sum of all the values is $\sum x = 59$. There are 11 numbers, so $n = 11$. Then the mean is

$$\bar{x} = \frac{59}{11} = 5.3636...$$

If there is only a finite range of values the data can take, it is easier to state the number of data points that take each value than to list them all. This kind of data is usually presented in a frequency table.

Example 2.2. Consider the same set of values from Example 2.1. We can present this same data in a frequency table.

The sum of the numbers in the frequency column is 11 as this is the total number of values.

To find the mean of the data in the table, we can multiply each by the frequency and add them together to get the value of $\sum x$.

$$(1 \times 1) + (1 \times 3) + (3 \times 4) + (2 \times 5) + (1 \times 7) + (1 \times 8) + (2 \times 9) = 59$$

This can be divided by the number of values (11) to get the mean of the dataset.

Value	Frequency
1	1
2	0
3	1
4	3
5	2
6	0
7	1
8	1
9	2

2.2.2 Grouped Data

When data points can take any value within a range, and there are too many values to list, we can use **grouped data**. This involves creating categories to split the data up, and then counting how many data points fall into each range.

When data is formatted in this way, we no longer have the true values of all the data points, but only which group they lie in. This means we can no longer get an exact value for the mean of this data - only an estimate.

Example 2.3.

Here is some data about the heights of children in a school year.

In order to estimate the mean, we need to find the midpoint of each range of values, and use this to represent that range. We can then treat it like a normal frequency table and calculate the mean.

Height (cm)	Midpoint	Frequency (f)
60-80	70	10
80-100	90	12
100-110	105	21
110-120	115	13
120-160	140	4

We multiply each midpoint value by its frequency and add them together to get the value of $\sum x$.

$$(70 \times 10) + (90 \times 12) + (105 \times 21) + (115 \times 13) + (140 \times 4) = 5536$$

The total number of data points is the sum of the frequency column: $10 + 12 + 21 + 13 + 4 = 60$

Then an estimate for the mean can be found by $\frac{\sum x}{n} = \frac{5536}{60} = 92.266\ldots \text{cm}$

The estimated mean height for a child in this group is $92.266\ldots \text{cm}$.

Exercise 2.1.

(a) Find the mean of these numbers:

77 431 875 716 843 601 482 325

(b) Find the mean of this frequency data:

x	0	1	2	3	4	5
Frequency	15	34	94	45	17	35

(c) Estimate the mean of this grouped data:

Class	Frequency
$0 \leq x < 10$	42
$10 \leq x < 20$	47
$20 \leq x < 30$	94
$30 \leq x < 50$	35
$50 \leq x \leq 100$	64

Function: AVERAGE() (E/S)

The AVERAGE() function (in Excel and Sheets) can be used to calculate the mean average of a data set. It takes as an input a range of values, and will ignore any non-numerical data included in the range.

2.2.3 Other Measures of Location

Definition 2.4 - Median

The **median** of a dataset is an average found by taking *the middle value* when the values are sorted into increasing numerical order. If there are an even number of values in the set, the median is given by the *mean of the two middle values* (if they are not the same). The median of a sample is sometimes denoted Q_2 .

Example 2.4. Consider the same set of values from Example 2.1:

4 9 7 5 1 8 3 4 9 5 4

Sorted into order, this set is:

1 3 4 4 4 5 5 7 8 9 9

Of the 11 values, the *sixth (middle) value* is *5* so this is the median of the data.

Function: MEDIAN() (E/S)

The MEDIAN() function (in Excel and Sheets) can be used to calculate the median average of a data set. It takes as an input a range of values, and will ignore any non-numerical data included in the range. If there are an even number of values in the set, it will calculate the mean average of the two in the middle.

Definition 2.5 - Mode

The **mode** of a dataset is an average found by taking *the value occurring most commonly in the data set*.
There is no standard symbol for the mode.

Example 2.5. For the dataset above, the value appearing most frequently is a *4* so this is the modal value.

(If the data is presented in a frequency table, finding the mode just involves *finding the largest number in the frequency column, and taking the corresponding value*.)

Not all datasets will have a unique mode; a data set with two modes is called *bimodal* and a data set with multiple modes more generally is called *multimodal*. It is also possible for a data set to have *no mode*, in the case where all values occur *once each*.

Function: MODE()/MODE.SNGL() and MODE.MULT() (E/S)

The MODE() or MODE.SNGL() function (in Excel and Sheets) can be used to calculate the mode of a data set. It takes as an input a range of values, and will ignore any non-numerical data included in the range.

The MODE.MULT() will return multiple values if the dataset is multimodal. This will output values in the cell it is entered in, and the cells below that if needed.

Definition 2.6 - Upper and Lower Quartiles

The upper and lower quartiles of a data set, denoted Q_1 and Q_3 are the values occurring one quarter and three quarters of the way through the set, when the values are sorted into increasing numerical order. They are not averages, but describe where certain areas of the data are.

Example 2.6. To find the upper and lower quartiles, we can split a data set in half using the median. Given our sorted data set from the Example 2.1:

1 3 4 4 4 5 5 7 8 9 9

The median was the first occurrence of 5 in the set.

We can split this into two sets (excluding the median itself), and find their medians:

1 3 4 4 4 5 7 8 9 9

Then we have $Q_1 = 4$ and $Q_3 = 8$.

(If there are an even number of values in a set, we would split it in half and find the median of each half.)

Example 2.7.

The table shows the number of kittens each of the children in a school year would like to own.

The column on the right shows the total number of kittens that the children would like to own. The mean (average) number of kittens each child wants can be calculated by dividing the total of this column (109)

by the number of children surveyed (60) giving $\bar{x} = \frac{109}{60} = 1.82...$

Kittens (x)	Frequency (f)	Total (xf)
0	10	0
1	12	12
2	21	42
3	13	39
4	4	16
Total	60	109

To find the median of this set, we could list all 60 values and find the middle value(s):

0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 3 3 3 3 3 3 3 3 3 3 3 4 4 4 4

In this case, the median value is 2. To find this from the table, we must first note that the total number of values is 60 so we are looking for the 30th & 31st values.

If the values are in order, the first ten will be 0, the next 12 after that are 1; $10+12=22$, but $10+12+21=43$ which is >30 . This indicates the median is 2, as the 30th & 31st values are both 2.

Similarly, we can determine that the quartiles will be the 14th/15th value and 44th/45th value so they will be 1 and 3.

Function: QUARTILES (E/S)

The QUARTILE function (in Excel and Sheets) can be used to calculate the quartiles of a data set. It takes two inputs: a range of values, and a number to indicate which quartile you want (0, 1, 2, 3 or 4). Choosing 0 will output the minimum value, 2 will output the median, and 4 will output the maximum value. It will ignore any non-numerical data included in the range.

Exercise 2.2.

- For each of the datasets in Exercise 2.1, find the mode (or modal category), and find or estimate the median and upper/lower quartiles.
- For the children's heights data from Example 2.3, find the modal category and an estimate of the upper and lower quartiles.

2.3 Measures of Spread

As well as measures of location, we can use statistics to describe the **spread** of a data set - describing how spread out the data are.

Definition 2.7 - Range

The range of a data set is given by the largest value in the set minus the smallest.

Example 2.8. In the example data set, the largest value is 9 and the smallest is 1 so the range is $9 - 1 = 8$.

Definition 2.8 - Interquartile Range

The interquartile range of a dataset is the difference between upper & lower quartiles.

Example 2.9. In our earlier example, we had $Q_1 = 4$, $Q_3 = 8$ Then the interquartile range, or IQR, is $8 - 4 = 4$.

Function: Range and IQR (E/S)

The MAX() and MIN() functions (in Excel and Sheets) can be used to find the maximum and minimum values in a dataset, and the difference between them (MAX() - MIN()) will tell you the range. You will need to give it a range of cells, which will all need to contain numerical values. (The MAXA() and MINA() functions will do the same thing, ignoring any non-numerical values in the range).

The IQR of a dataset can similarly be calculated using the QUARTILE() function, by calculating (e.g.) QUARTILE(A1:A10, 3) - QUARTILE(A1:A10, 1).

Exercise 2.3.

- For each of the datasets in Exercise 2.1, find or estimate the range and interquartile range.
- For the children's heights data from Example 2.3, using the quartiles you calculated earlier, find the interquartile range.

These measures tell us a little about how spread out the data are, but they don't use the whole data set, so can only capture some of the information. We have more powerful tools, which tell us more about the data.

2.3.1 Variance**Definition 2.9 - Variance**

The **variance** of a sample, denoted S_x^2 is a measure of how spread out the data are - *the larger the variance, the more spread out the data are.*

- and for a set of n values it can be calculated as:

$$S_x^2 = \frac{1}{n-1} \sum (x - \bar{x})^2$$

This formula says that for each data point in the sample, we find *the difference between it and the mean* then take the square of that value. Then we add these values together and divide by $n-1$ (so, if there are n values, we divide by the number one less than that).

Example 2.10. Consider the sample:

1 6 9 4 8 6

The mean of this set is $\bar{x} = \frac{(1+6+9+4+8+6)}{6} = 5.66\dots$

Subtract this number from each data point:

$$-4.66... \quad 0.33... \quad 3.33... \quad -1.66... \quad 2.33... \quad 0.33...$$

Then square each and take the sum:

$$(-4.66...)^2 + (0.33...)^2 + (3.33...)^2 + (-1.66...)^2 + (2.33...)^2 + (0.33...)^2 = 41.33...$$

Finally, divide by $n - 1$:

$$\frac{(41.33...)}{(6-1)} = 8.27...$$

So the variance is $S_x^2 = 8.27...$

Note

You might wonder why we square the differences from the mean in calculating the variance. If we were to just take the differences, some would be positive and others negative - and if we added them all together *we would get zero*. Squaring all the differences makes them *positive* and also improves the usefulness of variance as a measure.

We can also calculate the variance using a similar (but not the same!) formula:

$$S_x^2 = \frac{1}{n-1} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right)$$

Instead of calculating the differences between each value and the mean, we are now taking the difference between the *mean of the squares* and the *square of the mean*.

Example 2.11.

Consider the same sample as in Example 2.10.

1 6 9 4 8 6

The sum of the values is $\sum x = (1 + 6 + 9 + 4 + 8 + 6) = 34$. The sum of the squares is given by:

$$\sum x^2 = 1^2 + 6^2 + 9^2 + 4^2 + 8^2 + 6^2 = 1 + 36 + 81 + 16 + 64 + 36 = 234$$

So using the formula:

$$S_x^2 = \frac{1}{5} \left(234 - \frac{34^2}{6} \right) = 8.27...$$

This matches the previous value for the variance.

It's possible to show that these two formulae are equivalent. The second form is in general easier to calculate.

Exercise 2.4.

For each of the datasets in Exercise 2.1, find or estimate the variance.

Function: VAR.S (E) / VAR (S)

The VAR.S() function (in Excel) or the VAR() function (in Google Sheets) can be used to calculate the variance of a sample data set. It takes as an input a range of values, and will ignore any non-numerical data included in the range.

The variance does give a measure of how spread-out the data is, but it's less useful for meaningful comparisons. A more useful measure, and a fundamental idea in statistics, is the **standard deviation**.

2.3.2 Standard deviation

Definition 2.10 - Standard Deviation

The **standard deviation** of a sample, denoted s_x is the **square root of the variance** of that sample. It can be calculated as:

$$s_x = \sqrt{\frac{1}{n-1} \sum (x - \bar{x})^2} \quad \text{or} \quad \sqrt{\frac{1}{n-1} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right)}$$

Roughly, the standard deviation tells us the **average distance of each data point from the mean**.

Function: STDEV.S (E) or STDEV (S)

The STDEV.S() function (in Excel) and the STDEV() function (in Google Sheets) can be used to calculate the standard deviation of a sample. It takes as an input a range of values, and will ignore any non-numerical data included in the range.

Example 2.12.

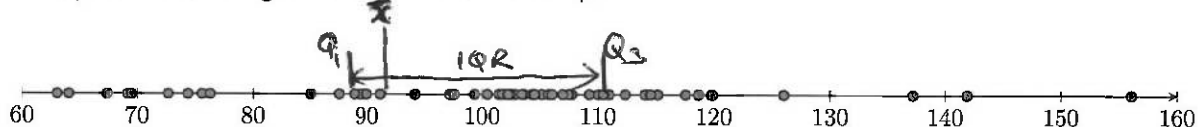
In Example 2.10, we calculated the variance of a sample to be 8.27. To find the standard deviation, we can take the square root to get $s_x = \sqrt{8.27} = 2.875$.

Exercise 2.5.

- (a) For each of the datasets in Exercise 2.1, find or estimate the standard deviation.
 (b) For the children's heights data from Example 2.3, find an estimate of the standard deviation.

2.3.3 Outliers

This plot shows the heights of the children from Example 2.3.



The mean of the data was estimated to be ~~92.2cm~~ ^{100.7cm} and we estimated the quartiles to be ~~89.9~~ ⁹⁰ and ~~110.5~~ ¹¹⁵.
 The IQR was estimated to be ~~20.6~~ ²⁵. The standard deviation of this data was ~~18.75~~ ^{18.42}.

While most of the data is clustered together in an area just higher than the mean, there are some data points which lie further away. Data points like this are called **outliers**, and many types of data will include them.

Definition 2.11 - Outliers

We can say a data point is an **outlier** if it lies more than $1.5 \times IQR$ or alternatively **3 standard deviations** away from the mean.

This is a widely-used definition, although it is not universal. Outliers can be due to false readings, and are sometimes excluded from datasets to facilitate analysis - although this should be done with care, especially if it might affect the validity of the results.

Exercise 2.6.

Which of the data points on the plot above are outliers, by the IQR definition, and by the standard deviation definition?

2.3.4 Limitations of Summary Statistics

For each of the measures we have seen, there are some advantages and disadvantages.

Mean, Standard Deviation and Variance These three numerical measures are commonly used for symmetrical data (data which is equally spread out around the middle), or for large sets of data (e.g. $n=1000$). Typically, the standard deviation and variance are used in conjunction with the mean.

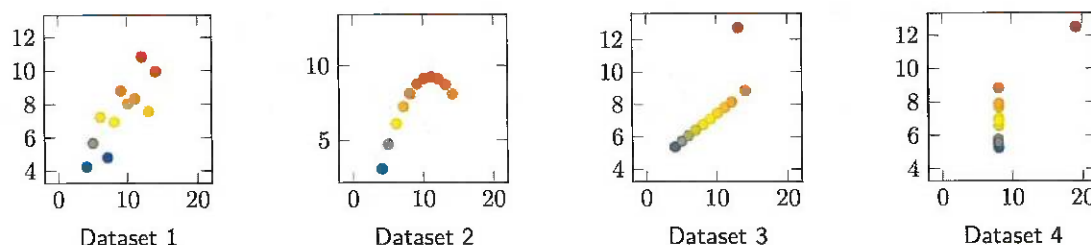
These measures use all the data values, giving a true indication of that measure, and each data value equally influences the measure being calculated. However, in a small sample, any extreme values may influence and skew the measure.

Median and Inter-quartile Range The median and interquartile range are commonly used with smaller sets of data, or data which are skewed to one side (e.g. more at the higher end than the lower end). Typically, the median is used in conjunction with the inter-quartile range.

They are not affected by extreme values, and can be calculated from incomplete data sets. However, these measures don't take into account all the values in the data. Without using technology, if the data set is large, the median may be difficult to locate.

Mode and Range The mode and range are commonly used when the data is incomplete or unknown. While they are both relatively easy to calculate, data sets may be multimodal, or have no mode - and depending on the context, the mode may not have any practical meaning. The range is also heavily influenced by extreme values.

2.3.5 Anscombe's Quartet



These plots show a collection of datasets known as **Anscombe's Quartet**. The summary statistics for these datasets are given in the table.

Even though the data sets look different, they have *the same mean and variance* in both the x and y directions. The data have been carefully chosen to have this property; it doesn't mean we shouldn't trust these measures, but we need to be careful not to rely on them too much.

Dataset	1	2	3	4
Mean x	9	9	9	9
Mean y (2 d.p.)	7.50	7.50	7.50	7.50
$s^2 x$	11	11	11	11
$s^2 y$ (± 0.003)	4.125	4.125	4.125	4.125

2.4 Populations and Samples

Definition 2.12 - Population and Sample

The **population** is the *entire body of subjects you are interested in*. Populations are generally too large to measure every subject, but are usually what we're interested in learning about. A **sample** is a smaller collection that is *a subset of the population*, which we can study in order to make inferences about the population.

Later we will see how to choose samples, and how to use the statistics we can calculate from the samples to infer things about the wider population.

The statistics we have seen use slightly different notation for the sample and the whole population.

Definition 2.13 - Notation for samples and populations

We use $\bar{x}$ to denote the sample mean, s_x^2 for the sample variance and s_x for the sample standard deviation.

When studying populations, it is more conventional to use the Greek letter μ (mu) to denote the population mean, and to write σ^2 for the population variance and σ for the population standard deviation. This is the Greek letter *sigma* which is the lower-case version of Σ .

The calculation for the variance and standard deviation of a population is slightly different - instead of dividing by $n - 1$, we divide by n , so the formulae become:

$$\sigma^2 = \frac{1}{n} \sum (x - \bar{x})^2 \quad \sigma = \sqrt{\frac{1}{n} \sum (x - \bar{x})^2}$$

Function: VAR.P and STDEV.P (E) / VARP and STDEVP (S)

The VAR.P() function (in Excel) or the VARP() function (in Google Sheets) can be used to calculate the variance of a population data set. It takes as an input a range of values, and will ignore any non-numerical data included in the range.

The STDEV.P() function (in Excel) or the STDEVP() function (in Google Sheets) can be used to calculate the standard deviation of a population data set. It takes as an input a range of values, and will ignore any non-numerical data included in the range.